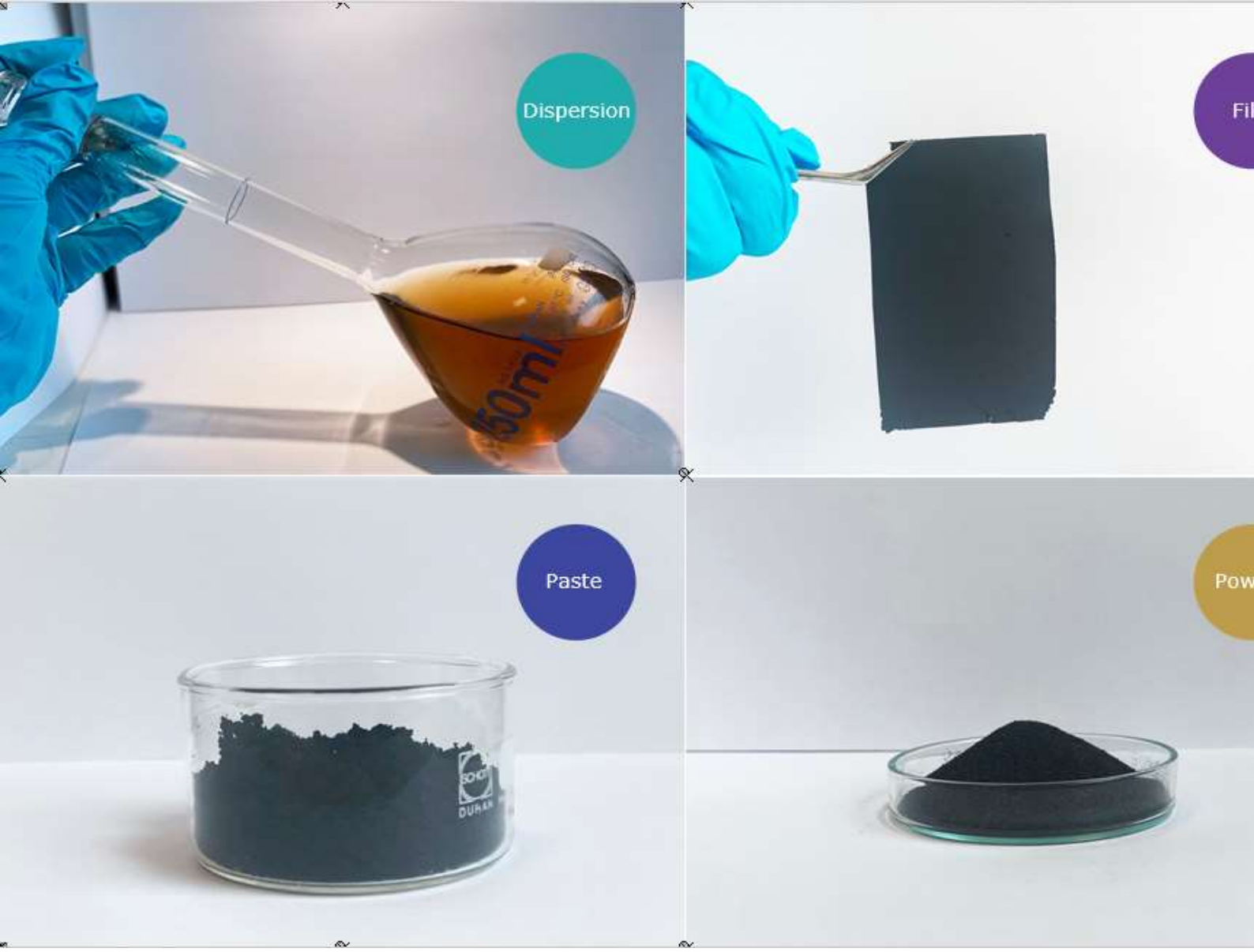
Nordische

Active Materials

Version 1.6

2023



We create industry-friendly technologies that enable the use of graphene materials in a vast variety of applications and industries. Our core focus is on energy storage applications, coatings & composites and advanced water solutions

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Learn more about different technologies from our website
www.nordischeenergy.com

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CARBONOUS PRODUCTS

Graphene A+ (Battery Grade)

High surface area graphene with excellent electrical and thermal conductivity, nanoparticles thickness go down to single–layer graphene. It is easily dispersible in various battery solvents such as NMP.

Thickness: < 0.8-1.6nm , Purity: > 99%

PRODUCT FEATURES

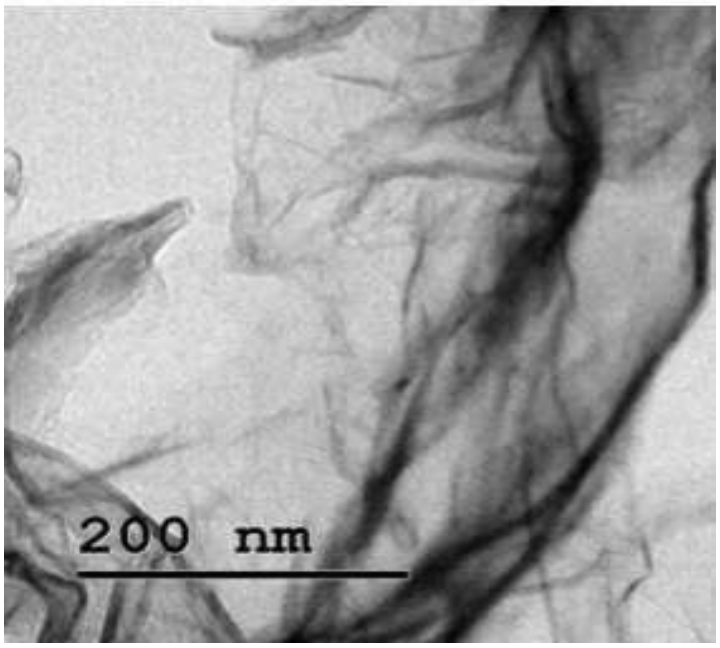
- It has exceptional electrical conductivity, allowing efficient electron transport within the battery system, as a result, fast charging batteries can be developed
- The increased surface area provides more active sites for electrochemical reactions, facilitating higher energy and power densities, energy conversion, optoelectronic devices and catalyst
- Its high surface-to-volume ratio and ability to accommodate ions on its surface making it an ideal material for enhancing the capacity of batteries, leading to long-lasting power sources
- It can withstand repeated charge and discharge cycles without significant performance degradation, resulting in a longer battery lifespan

Applications

Lithium-ion Batteries (Li-ion)	Sodium-ion Batteries (Na-ion)	Aluminium-ion Batteries (Al-Ion)
Zinc-Ion Batteries (Zn-Ion)	Lead Acid Batteries	Super Capacitor

Specifications

Ng A +	Description
Purity	99.9%
Primary Lateral Dimension	< 1 µ
Particle Size	D50: < 30 µ
Thickness	0 . 8 – 1 . 6 Nm
Bulk Density	0. 1 3 - 0 . 1 4 G / Cm3
Physical Form	Pluffy Powder
Colour	Black
Surface Area	~ 4 0 0 M2/G
CAS No..	7782 - 42 – 5



Graphene D (Industrial Grade)

We Produce industrial grade Graphene (NADG-D) nanoplatelets in bulk quantity by chemical exfoliation methods and further treat it for purification with multiple quality checks.

Thickness: <5-10nm , Purity: > 99%

NG-D

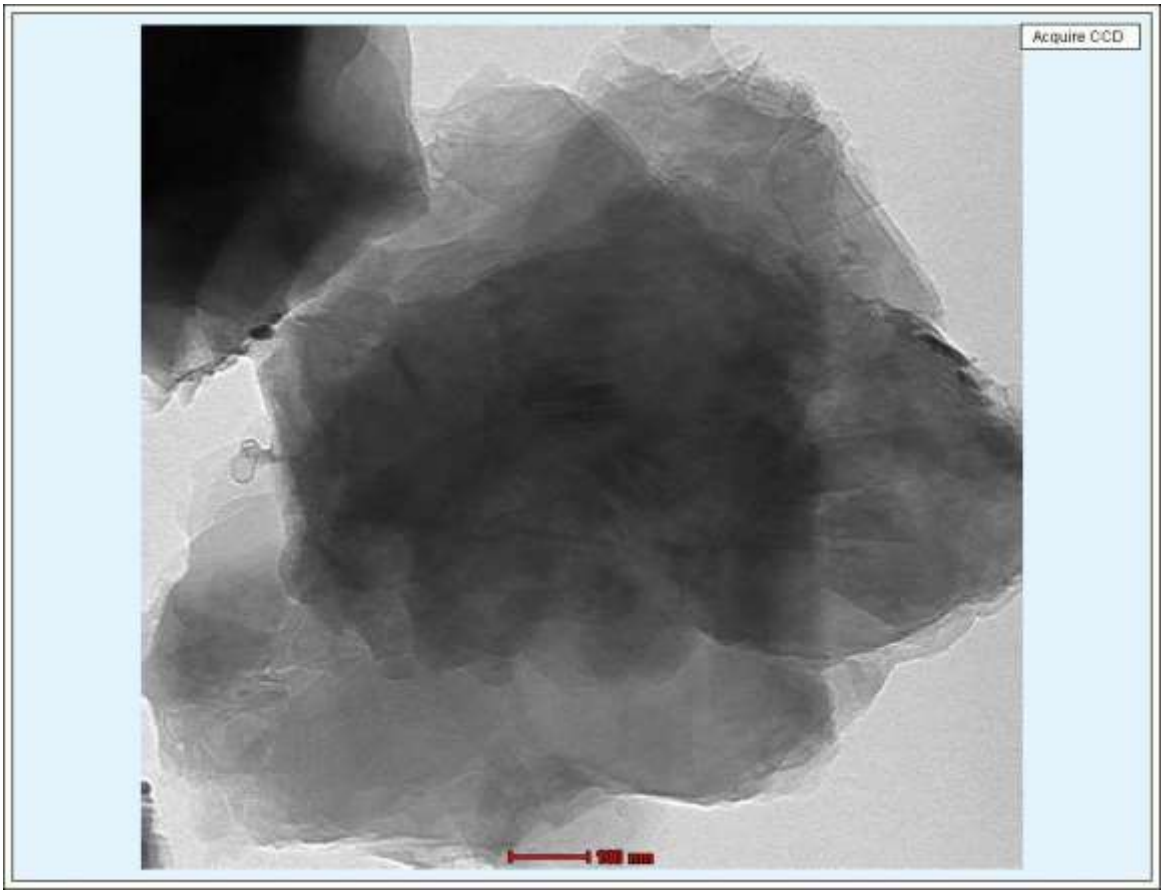
PRODUCT FEATURES

- Improves Electrical, Thermal conductivity and Mechanical stability
- Used to improve the properties of Polymer
- Used to improve the properties of Paints and Coatings
- Good Mechanical, Electrical, and Thermal properties
- Small addition will improve the properties of Matrix

Applications

Used to improve the properties of polymer composites	Used to improve the properties of Thermoplastics	Used to improve the properties of Tyres
Used to make Advance concrete Anti-corrosion	Used to improve the properties of Automobile components	Used in Paints & Coatings, Solar Panels, Dispersion & Inks

Graphene D(NADG -D)	Description
Purity	99.9%
Particle Size	D 5 0 : <1 - 2 µ
Thickness	5 - 10 Nm
Bulk Density	0. 166 G / Cm3
Physical Form	Powder
Colour	Greyish Black
Surface Area	~ 11 0 M2/G
CAS No. .	7782 - 42 – 5



Graphene P (Paints and Coating Grade)

When this grade is incorporated into the primer with less zinc content, it forms a dense and impermeable barrier that prevents moisture, oxygen and corrosive agents from reaching the substrate. This barrier protection significantly reduces the risk of corrosion and extends the life span of the coated surface and the environmental friendly coating.

Thickness: <1-5 nm , Purity: > 99%

PRODUCT FEATURES

- Its excellent impermeability to gases, liquids, and corrosive substances makes it an ideal candidate for enhancing the barrier properties of anti-corrosion coatings
- It helps in providing superior protection against corrosion, extending the lifespan of ships, offshore platforms, buoys, and other marine structures
- Graphene's strong bonding properties enable improved adhesion between the coating and the substrate, ensuring better protection against corrosion
- Graphene can enhance the interfacial adhesion strength and reduce the risk of delamination or detachment of the coating, which can be a common cause of corrosion initiation
- It facilitates the flow of electrons and enhances the efficiency of cathodic protection, thereby minimizing the corrosion rate on metal surfaces

Applications

Infrastructure Protection	Marine and Offshore	Industrial Equipment
Aerospace and Aviation	Oil and Gas Industry	Automotive Industry

Graphene P (Nadg-P)	Description
Purity	99.9 %
Particle Size	D50 :< 1.0 µ
Thickness	1-5 Nm
Bulk Density	0.2 G / Cm3
Physical Form	Powder
Colour	Black
Surface Area	~200m2 / G
CAS No. .	7782 - 42 - 5

Graphene Oxide

We make Graphene Oxide (GO) from modified Hummers Method. GO is an oxidised form of graphene, laced with oxygen-containing groups. It has been purified with multi levels of quality checks to obtain phase pure of GO.

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Thick: 0.8-2nm , Purity: > 99%

PRODUCT FEATURES

- High aspect ratio, very thin exfoliated sheet
- It is electrically insulating due to disruptive sp2 bonding network but it can be converted conductive by restoring network
- Easy dispersibility in water and other organic solvents
- Improves mechanical and barrier properties

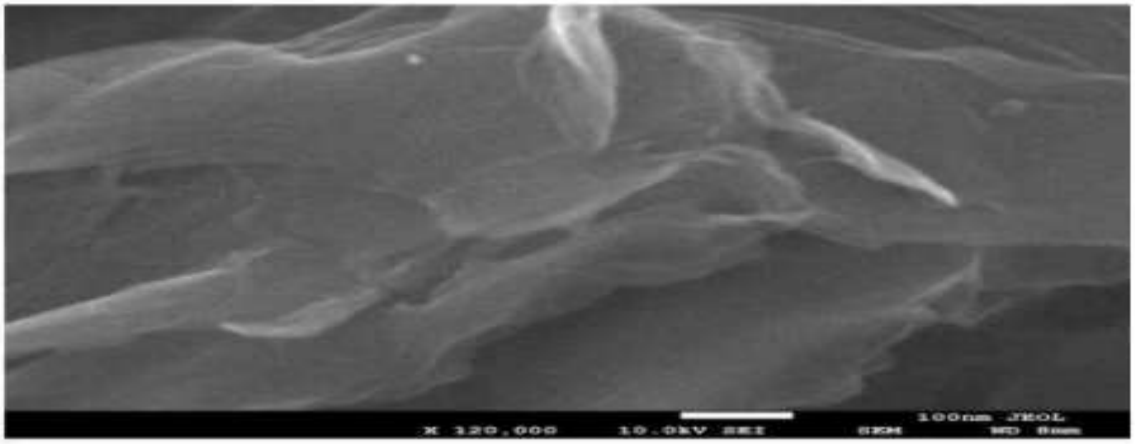
Applications

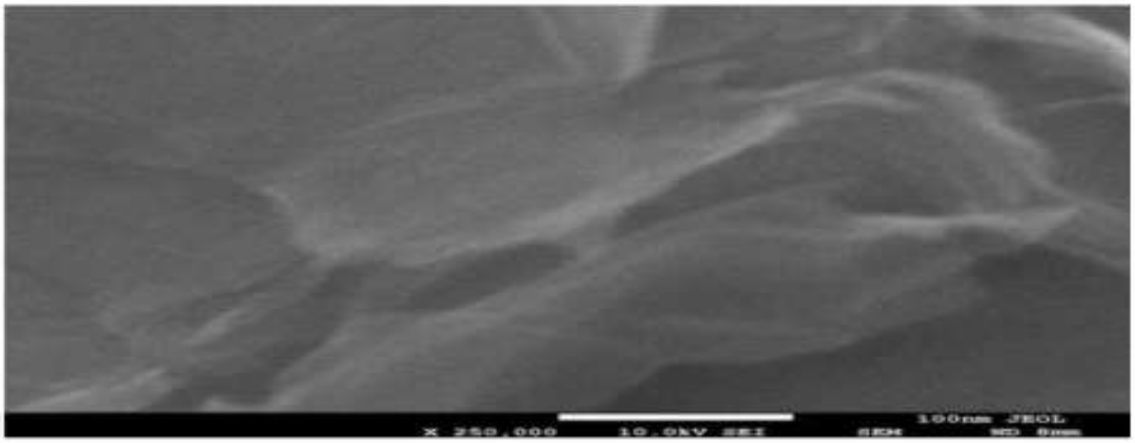
Used to Improve the properties of concrete	Used for Water Desalination	Used in solar cells
Used in Flexible Electronics	Used To Make Anti-barrier Coatings	Used for chemical sensors
Used to improve the Properties of polymers.	Used To Improve Properties of Elastomers	Small Quantity Can Improve Mechanical, Electrical Properties Of Paints & Coatings

Specifications

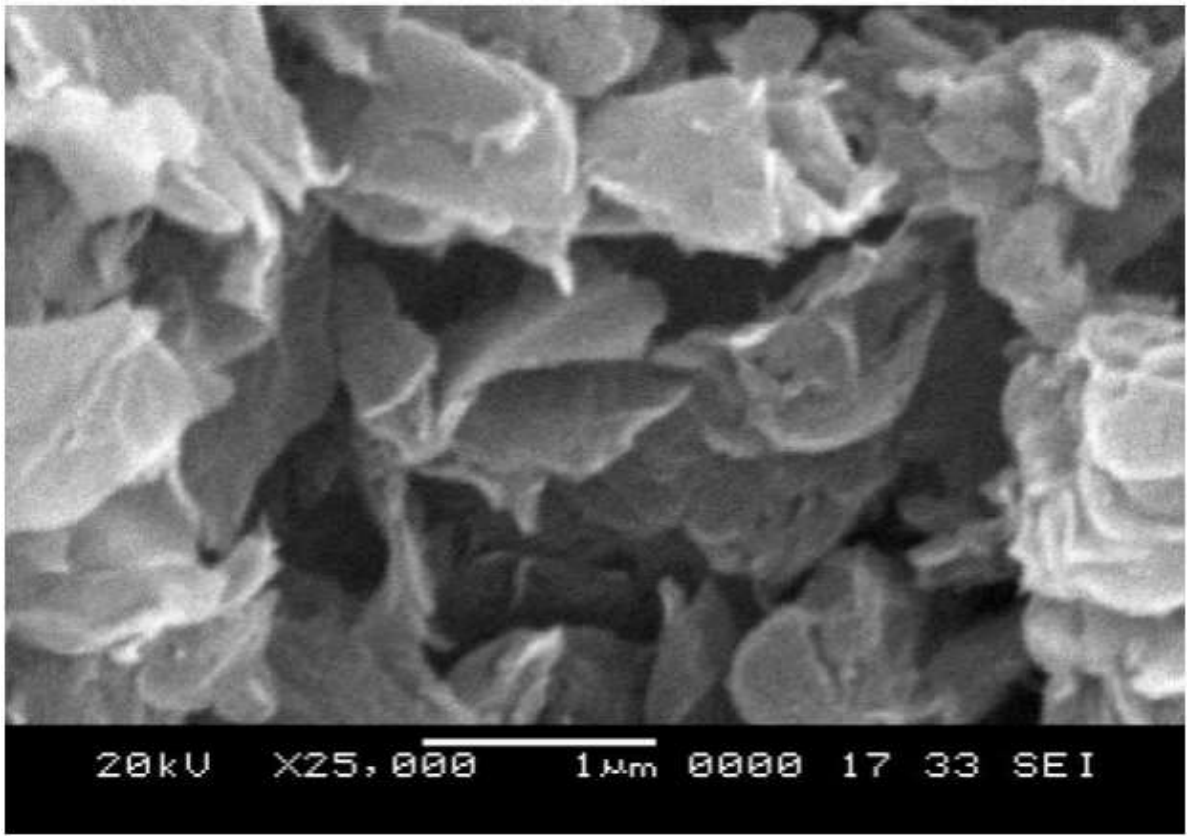
Reduce Graphene Oxide	Description
Purity	~99%
Thickness (Z)	~0.8 - 2 Nm
Average Lateral Dimension(X&Y)	~ 5 - 10 µ M
Number of Layers	1 – 3 Layers
Carbon Content	~60 - 80%
Oxygen Content	~15 - 32%
Hydrogen	~1 - 2%
Nitrogen Content	~1 - 2%
Sulfur Content	<1%
Surface Area	110 - 250* M 2 /G
Bulk Density	0. 05 G/Cm 3
CAS No..	7782 - 42 - 5
Physical Form	Fluffy, Very Light Powder
Odour	Odourless
Colour	Black Powder

FESEM Results

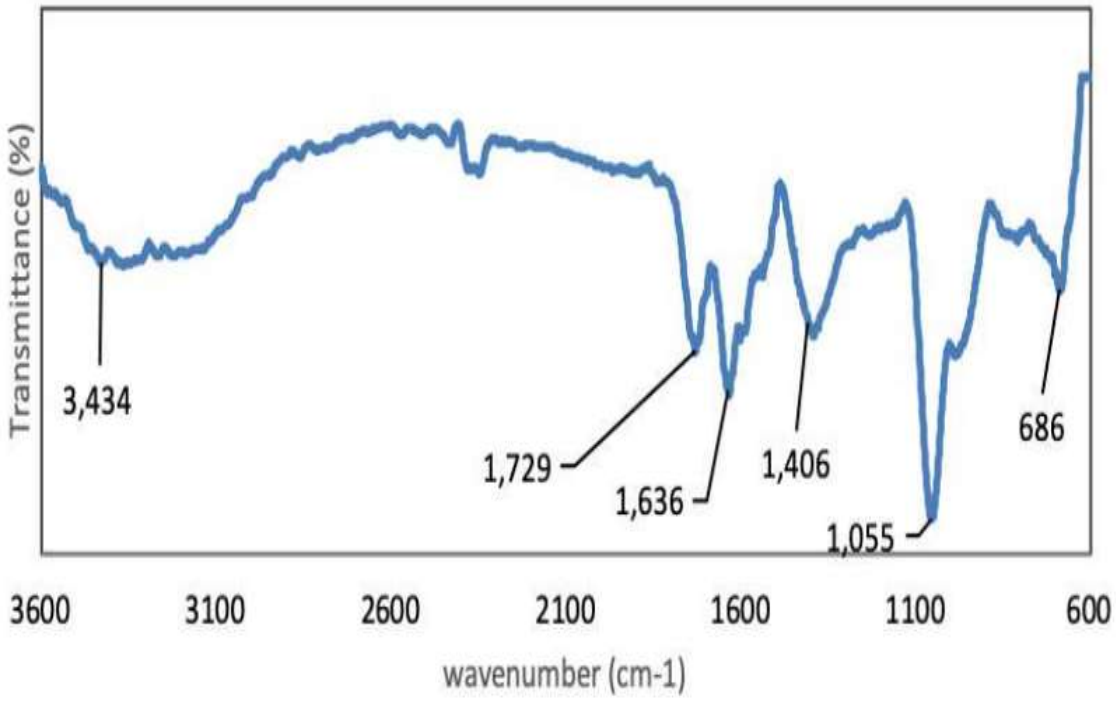




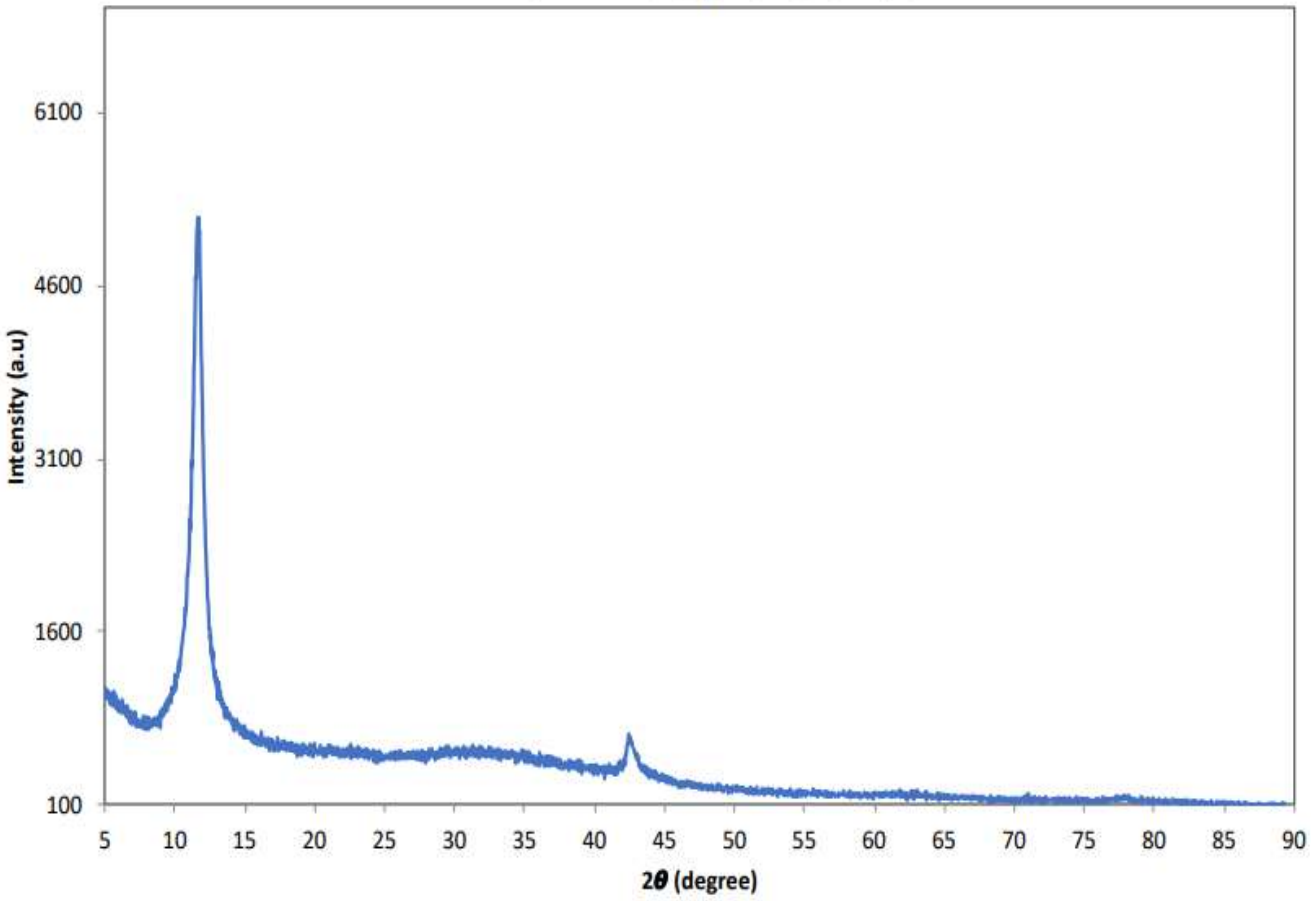
SEM Results



FTIR Results



XRD Results


6100
4600
3100
1600
100
Intensity (a.u)
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
2θ (degree)

THICK:0.8-2mm , Purity: > 99%

Reduced Graphene Oxide

We produce Reduced Graphene Oxide (RGO) by the reduction of Graphene Oxide (GO) , it has to be taken into account once most of the oxygen groups are removed.

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PRODUCT FEATURES

- High aspect ratio, Very thin exfoliated sheet
- Ultimately High Purity
- Improves mechanical Properties.
- Improve Electrical Properties.
- Enhance Thermal Conductivity.
- Small Addition will improves the properties of matrix

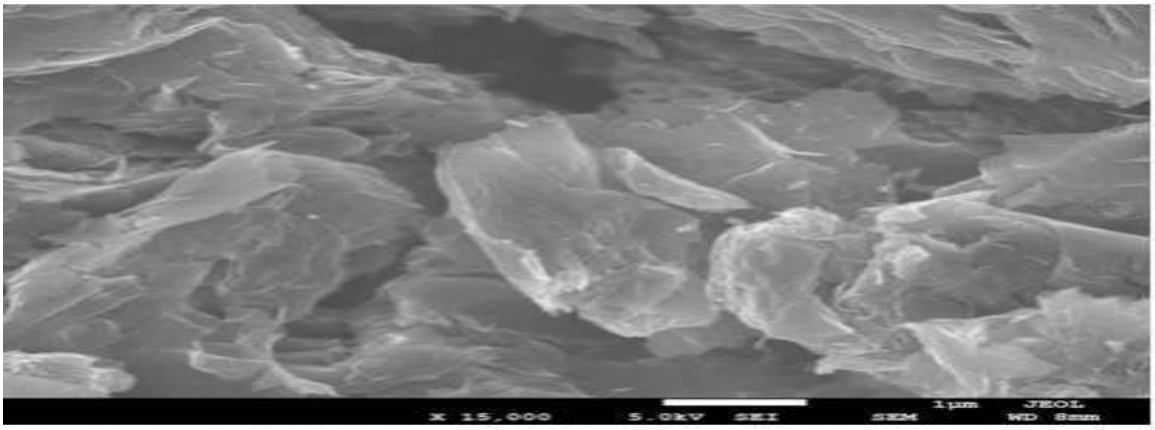
Applications

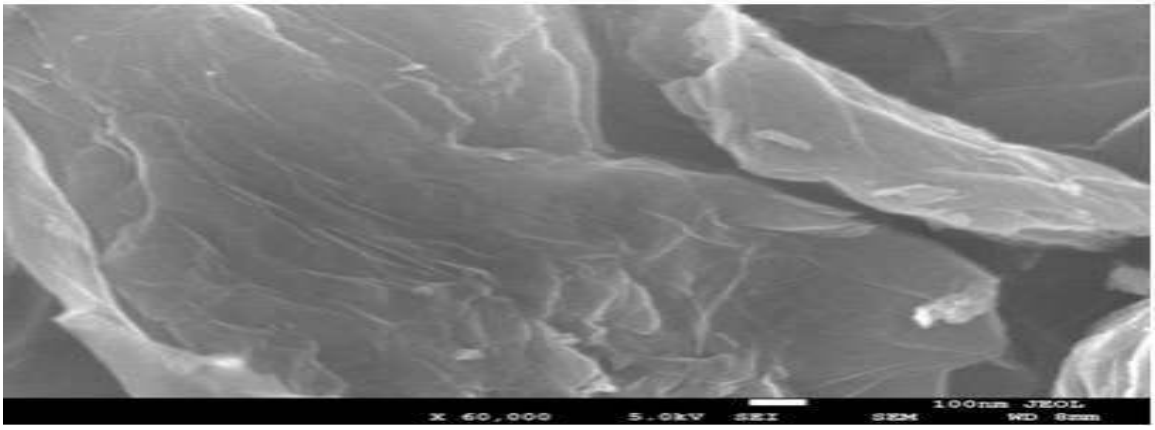
Small Quantity Used To Make Anti-corrosion Coatings	Used To Improve The Properties Of Polymer	Improve Power And Energy Density And Also Extend Battery Cycle Life
Used To Make Chemical Sensors	Used To Make Anti-barrier Coatings	Very Minute Quantity Can Improve Properties Of Concrete
Used To Improve The Properties Of Elastomers	Used To Make RFID Coatings	Small Quantity Can Improve Mechanical, Electrical Properties Of Paints & Coatings

Specifications

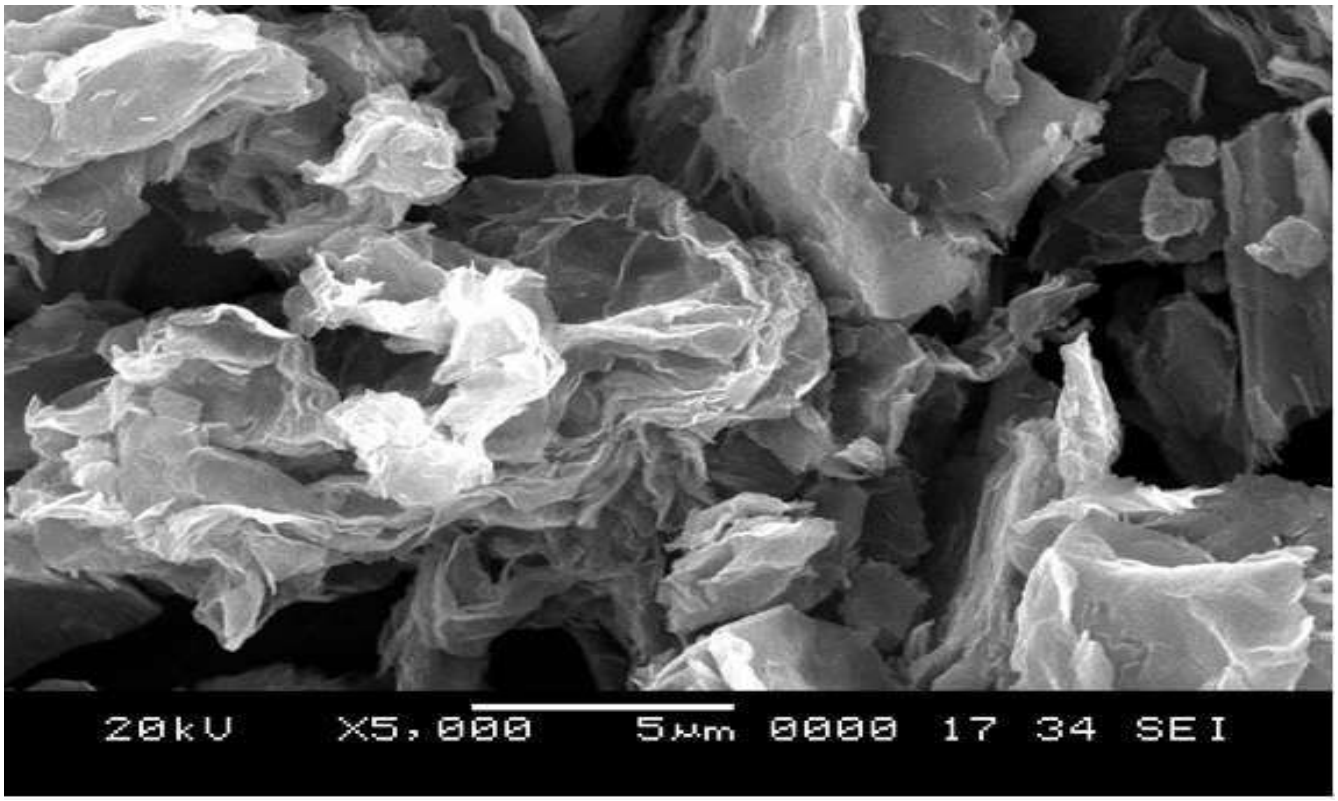
REDUCE GRAPHENEOXIDE	DESCRIPTION
Purity	99%
Thickness	0.8 - 2 nm
Average Lateral Dimension(X&Y)	5 - 10 µ m
Number of Layers	1 - 3
Carbon Content	~85 - 92%
Oxygen Content	~5 - 10%
Hydrogen	1 - 2%
Nitrogen Content	1 - 2%
Sulfur Content	<1%
Surface Area	80 - 200* m 2 /g
Bulk Density	0. 03 g/cm 3
CAS No..	7782 - 42 - 5
Odour	Odourless
Colour	Black Powder

FESEM Results

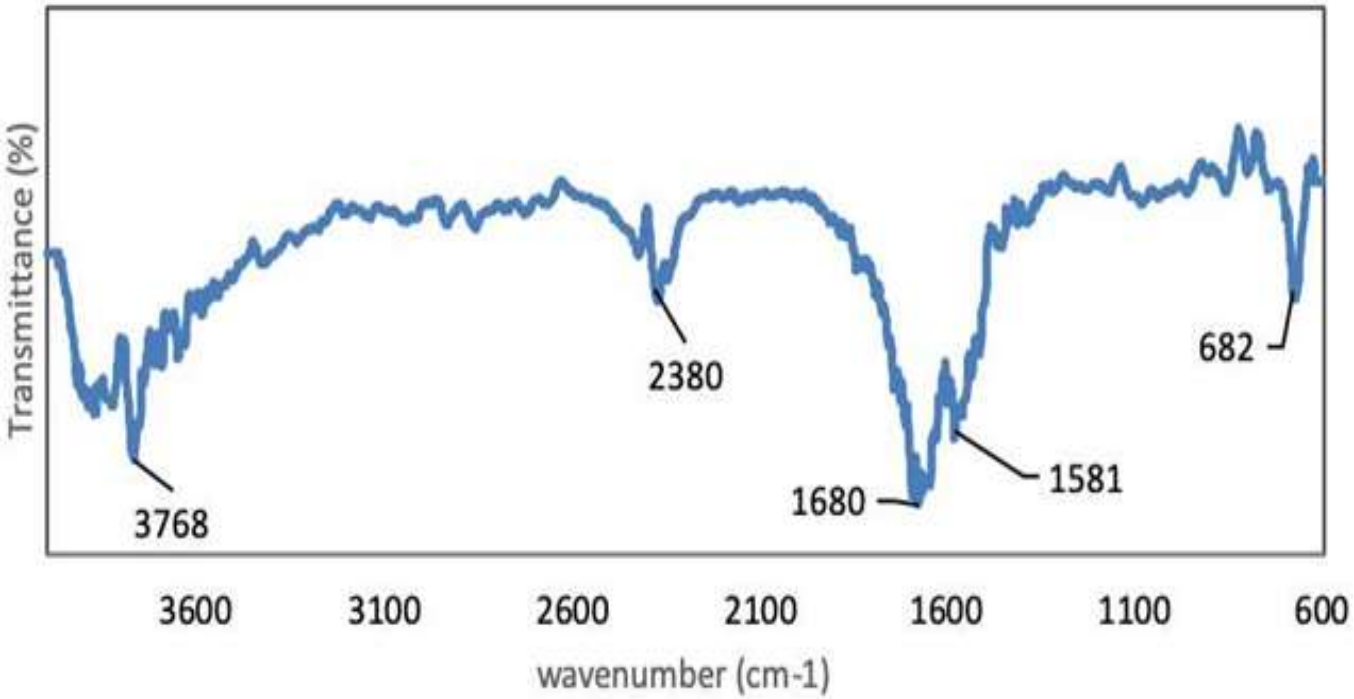




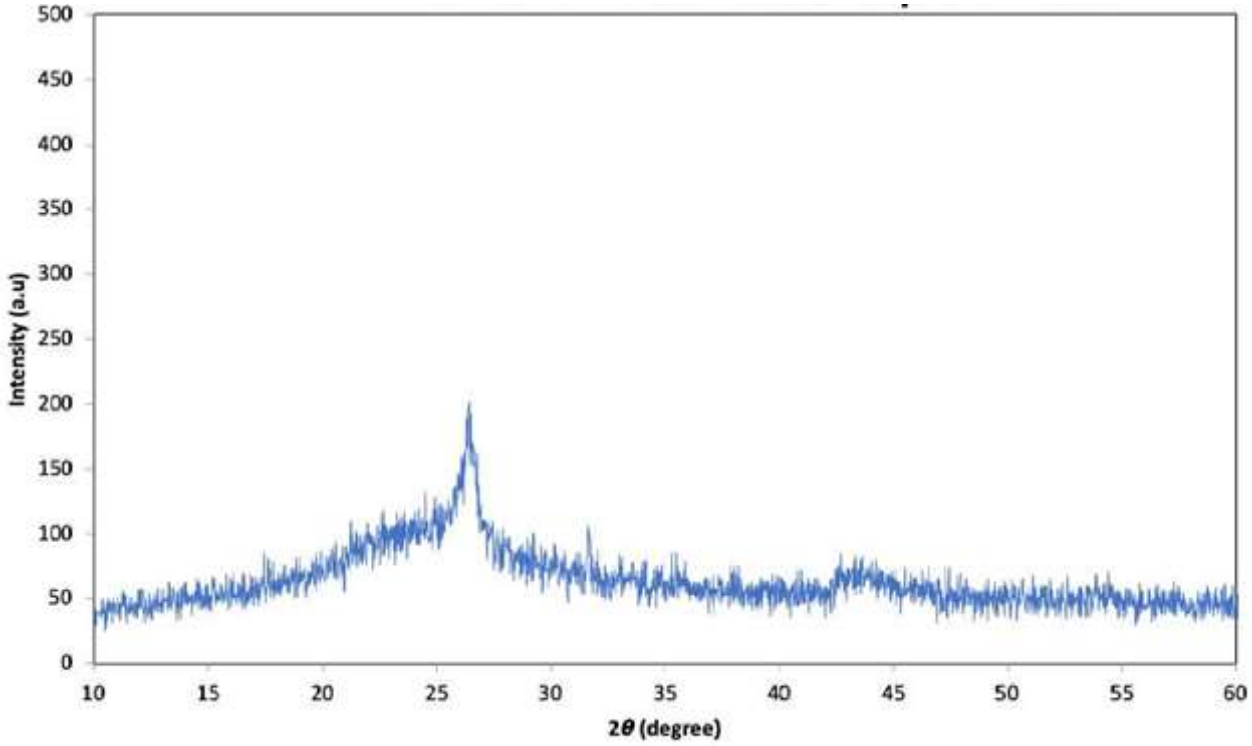
SEM Results



FTIR Results



XRD Results


Intensity (a.u)
500
450
400
350
300
250
200
150
100
50
0
10
15
20
25
30
35
40
45
50
55
60
2θ (degree)

Multi Walled Carbon Nanotube (NAM-5)

MWCNT are prepared by the low-cost bio fuel combustion method followed by purification.

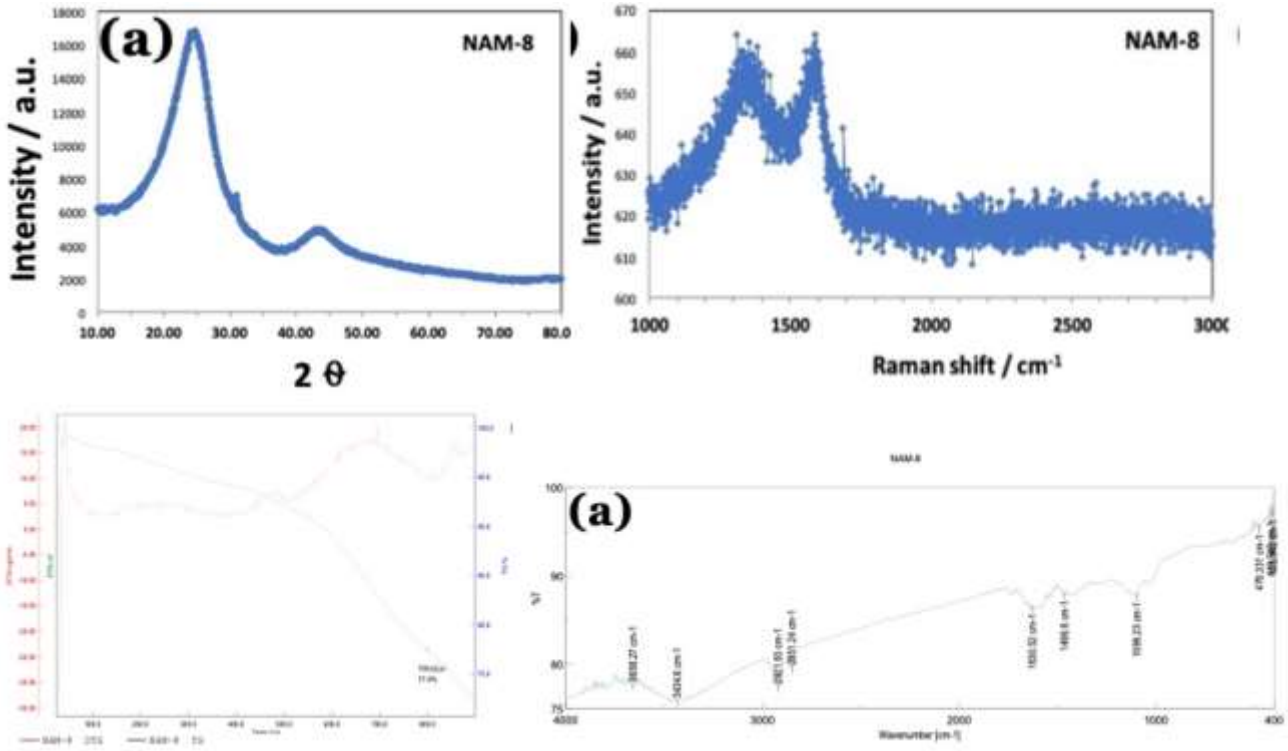
MWCNT (NAM-5)

MWCNT	DESCRIPTION
Ingredients	Carbon
Purity	99%
Diameter	10-20nm
Average Lateral Dimension(X&Y)	~10 µm
CAS No..	7782-42-5
Odour	**Odourless**
Colour	**Black Powder**

There are a number of advantages of this product and they are:

- High Aspect Ratio
- Dispersed much easier
- Small addition will improve the properties of matrix for improves electrically conductive.
- Improves thermal conductivity
- Improves mechanically stability

XRD, DTG and



XRD, TG and DTG, Raman – profile of MWCNT.

Applications

Small enforcement can improve the properties of polymers.

Improve mechanical, electrical properties of paints & coatings

Used to make sensors, fuel cells, emi shield coatings, highly conductive ink.

Used to improve the properties of thermoset, thermoplastics

Safety and Storage

- Double sealed, moisture free, plastic bottle,
- Store in dry and room temperature.

Acid Functionalized MWCNT (NAM-9-1)

Multi Walled Carbon Nanotubes from (MWCNT) is a highly porous material and has a hollow structure, high surface area, thermal stability, good thermal and electrical conductivity and optical activity, as well as mechanical damage resistance. Acid group functionalization using $HNO_3+H_2SO_4$ has been done to MWCNT on which further treatment is done to introduce -OH group at the edges and surfaces with multiple quality checks.

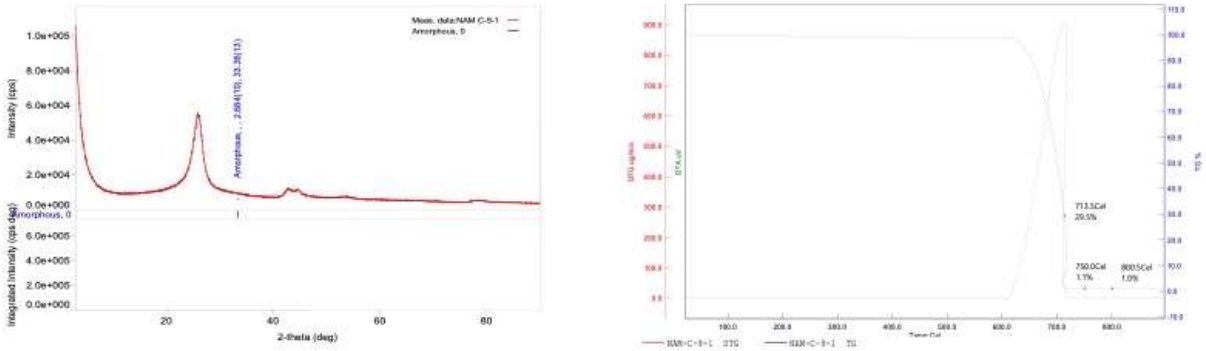
Acid Functionalized MWCNT (NAM-9-1)

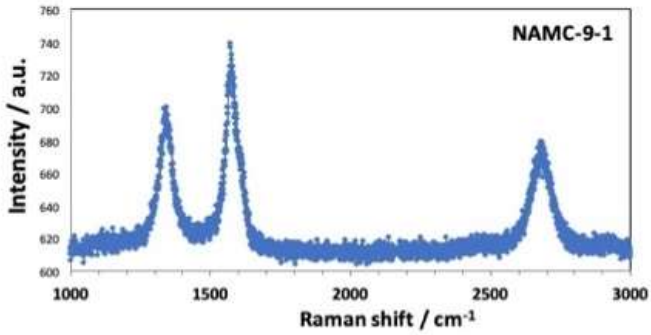
ACID FUNCTIONALIZED MWCNT	DESCRIPTION
Ingredients	**Carbon**
Purity	**99%**
Diameter	**10-20nm**
Average Lateral Dimension(X&Y)	**~10 µm**
CAS No..	**7782 - 42 - 5**
Odour	**Odourless**
Colour	**Black Powder**

There are a number of advantages of this product and they are:

- Dispersed much easier

- It has excellent dispersion and compatibility with polar solvents
- Small Addition will improve the properties of matrix 4 Improves electrically conductive
- Improves thermal conductive
- Improves mechanically stability





XRD, TG and DTG, Raman – profile of Acid functionalized MWCNT.

The key application areas of Hydroxyl functional MWCNTs are Solar cells, Batteries, Nano-electronics, Transistors, Energy storage, and Flat Panel Display., thermoplastic and thermoset composites, characteristic or engineered elastic, thermoplastic elastomers, cements, paints and coatings.

Safety and Storage

- Double sealed, moisture free, plastic bottle,
- Store in dry and room temperature.

Carboxylic Functionalized MWCNT (NAM-9-3-A)

Carboxylic group functionalization to MWCNT on which further treatment is done to introduce -COOH group at the edges and surfaces with multiple quality checks.

Carboxylic Functionalized MWCNT (NAM-9-3-A)

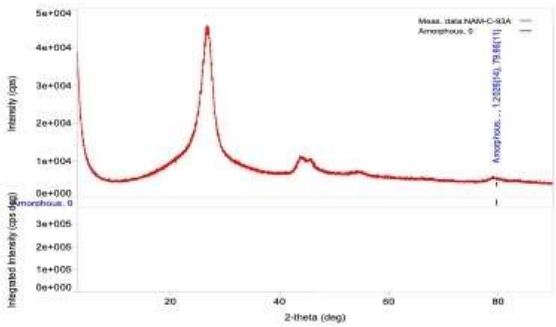
Carboxylic Functionalized Mwcnt	Description
Ingredients	Carbon
Purity	99%
Diameter	10-20nm
Average Lateral Dimension(X&Y)	~10 µm
CAS No..	7782 - 42 - 5
Odour	**Odourless**
Colour	**Black Powder**

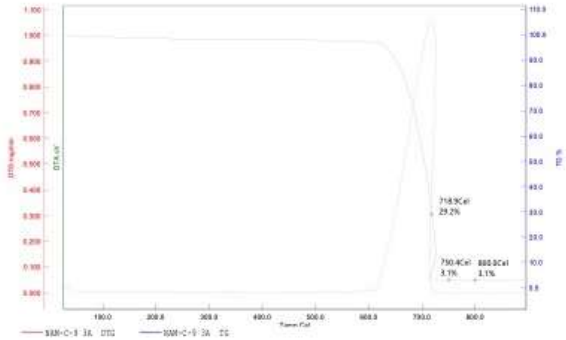
There are a number of advantages of this product and they are:

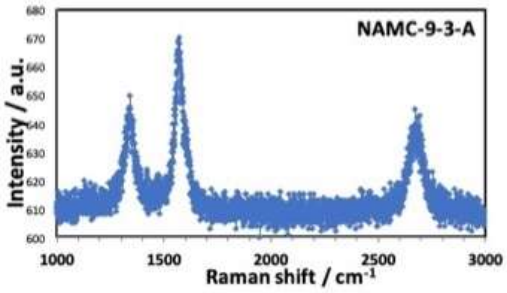
- High Aspect Ratio
- Dispersed much easier
- Carboxyl groups are compatibility with a number of polymers, including alkene and resin chemistries and much more
- Small addition will improve the properties of matrix for improves electrically conductive.
- Improves thermal conductivity

XRD, DTG and Raman

- Improves mechanically stability







XRD, TG and DTG, Raman – profile of Carboxylic functionalized MWCNT.

Applications

Small enforcement can improve the properties of polymers.	Improve mechanical, electrical properties of paints & coatings
Used to make sensors, fuel cells, emi shield coatings, highly conductive ink.	Used to improve the properties of thermoset, thermoplastics

Safety and Storage

- Double sealed, moisture free, plastic bottle,
- Store in dry and room temperature.

OXIDE PRODUCTS

Aluminium Oxide Nanoparticles

We manufacture Al2O3 nanoparticles by Sol-Gel method in bulk quantity with multi levels of quality checks to get pure phase Aluminium oxide Nanoparticles.

PRODUCT FEATURES

- High thermal barrier properties as well as Fire Retardancy.
- Durability, Stiffness, Thermal fatigue resistance, Fracture toughness, Creep resistance and wear resistance.
- Chemically stable
- Anti-corrosive
- Mechanical strength Enhancement
- Electrical insulating

Applications

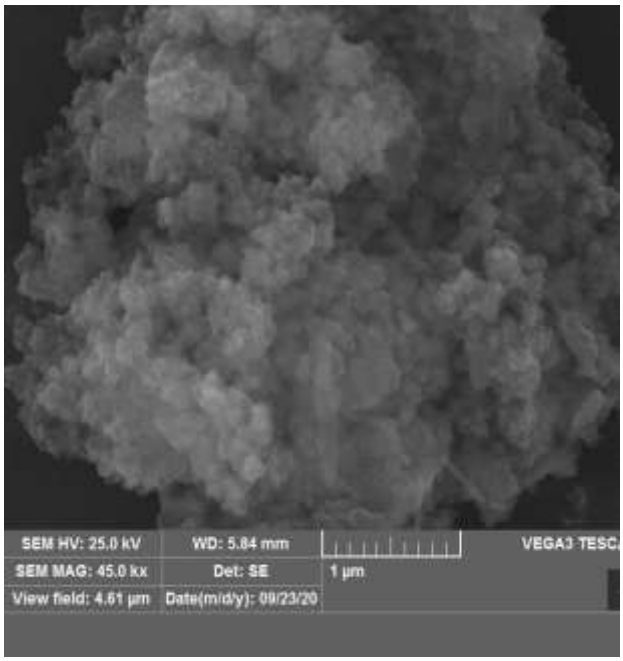
- Paints & Coatings
- Polymer Nano Composite
- Ceramics
- Automotive Industries
- Oil and Gas

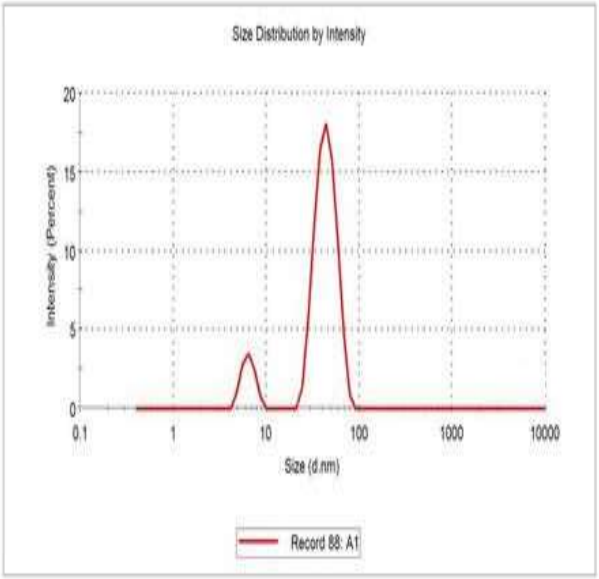
Specifications

Aluminium Oxide Nano Particle Powder

Al 2 O 3	Description
Purity	99.9 %
Average Particle Size	< 100 Nm
Bulk Density	0 . 5 G/Cm3
Ssa	110 M2/G
Physical Form	Powder
Morphology of Nano Particle	Near Spherical
Colour	White
CAS No.	1344–28-1

SEM RESULTS





Copper Oxide Nanoparticles

We manufacture Copper oxide (CuO) nano particles by Sol-Gel method with multi levelled quality checks to obtain phase pure Copper Oxide Nanoparticles.

APS: <100nm , Purity: > 99%

PRODUCT FEATURES

- Antimicrobial activity.
- Applicable for Energy conversion, Optoelectronic devices and catalyst.
- Super thermal conductivity, Photovoltaic properties, High stability
- Have Semiconducting Property
- Useful in batteries, catalysis, solar energy conversion, high-temperature superconductors

Applications

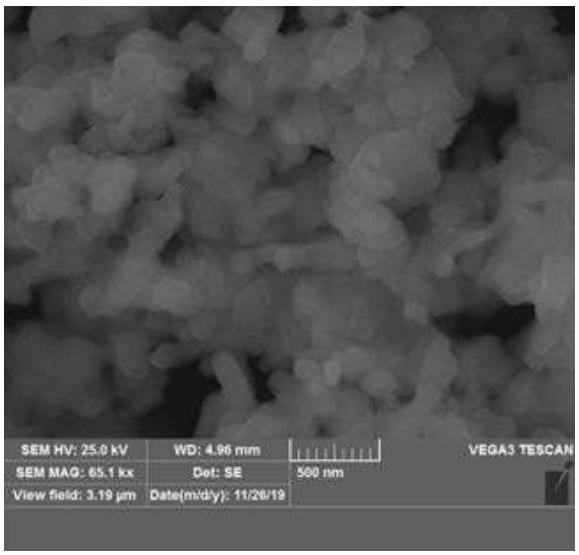
Solar	Antimicrobial	Sensors
Batteries	Electronics	Agricultural Industries

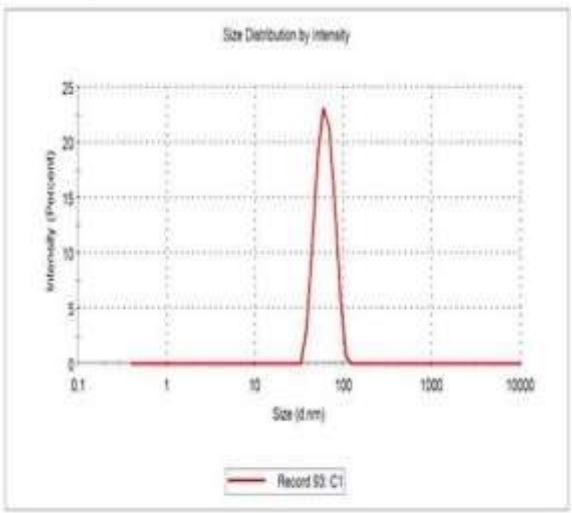
Specifications

Copper Oxide Nanoparticles

Cuo	Description
Purity	99.9 %
Average Particle Size	< 100 Nm
Bulk Density	0 . 66 G / Cm3
Ssa	60-80 M2/G
Physical Form	Powder
Morphology Of Nano Particle	Near Spherical
Colour	Dark Brown
CAS No.	1314-13-2

SEM RESULTS





Magnesium Oxide Nanoparticles

We manufacture MgO nanoparticles by sol-gel method in bulk quantity with multi levels of quality checks to obtain pure phase Magnesium Oxide nanoparticles.

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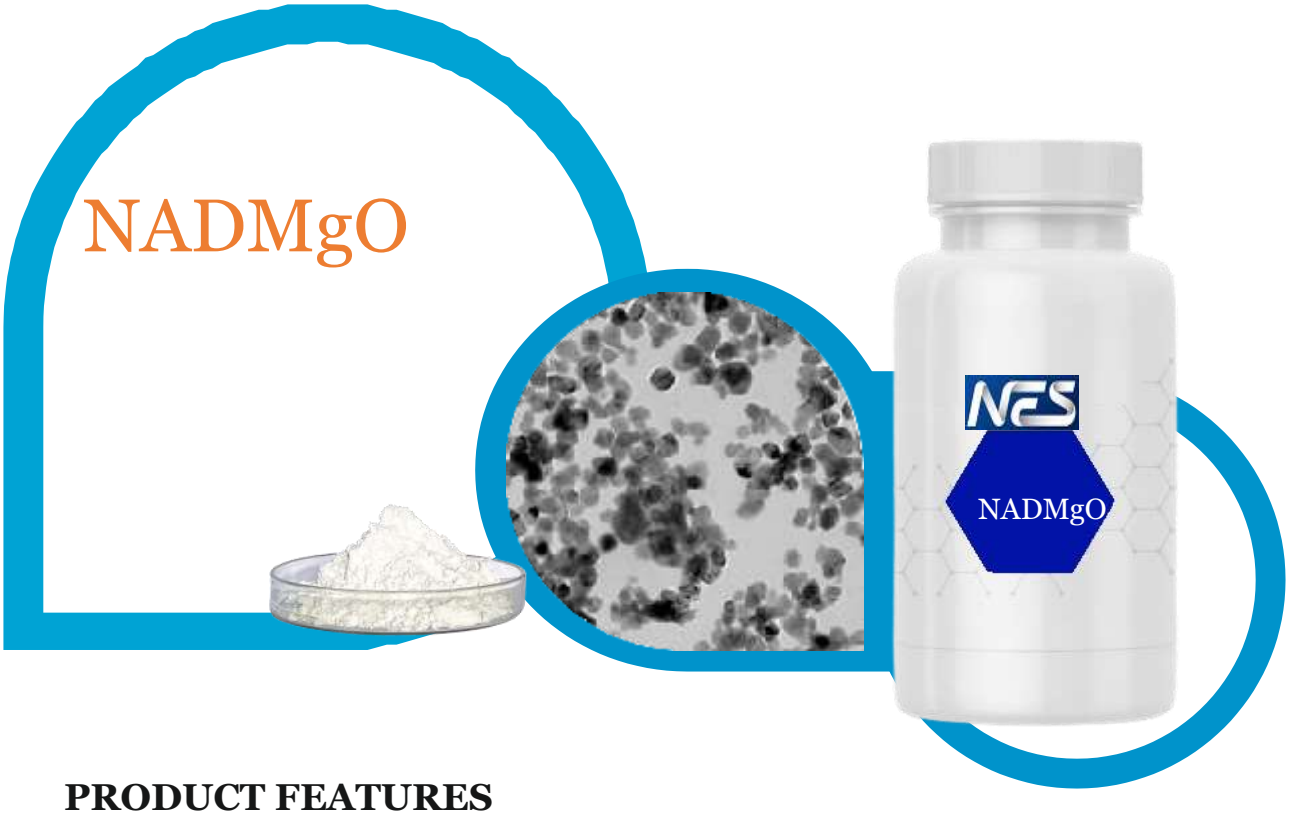


APS: <100nm , Purity: > 99%

PRODUCT FEATURES

- Magnesium oxide nanoparticles are non-toxic
- They possess high hardness, high purity
- It has good antibacterial property
- Anti Corrosive Properties
- High Thermal Properties
- Electrical insulator and Fire Retardancy

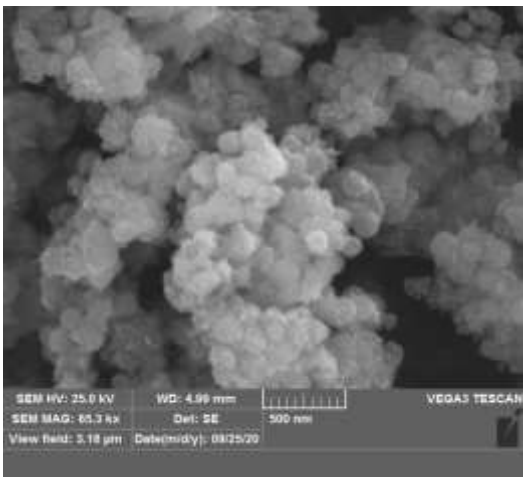
Applications

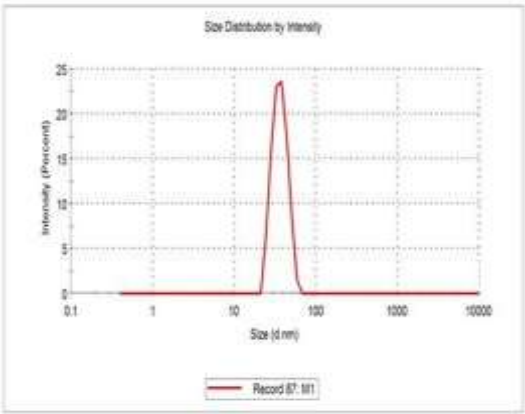
Cosmetics	Ceramic Coating	Oil and Gas
Paints and Coatings	Polymer nano composite	Automotive Industries

Specifications

MgO	DESCRIPTION
Purity	99.9 %
Average Particle Size	<100nm
Bulk Density	0.6 G / Cm3
Physical Form	Powder
Colour	White
Surface Area	110~130m2 / G
CAS No. .	1309-48-4
Morphology Nano Particle	Near Spherical

SEM RESULTS





Silicon Dioxide Nanoparticles

We manufacture SiO2 nanoparticles by sol-gel method in bulk quantity with multi levels of quality checks to obtain pure phase Silicon Dioxide nanoparticles.

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APS: <100nm , Purity: > 99%

$NADSiO_2$

PRODUCT FEATURES

- High thermal conductivity. Exceptionally low moisture absorption
- Extremely low permeability to gases and to ionic contaminants
- No Thermal wells or Thermal vias. Supports built-in thin film terminating resistors.
- Anti microbial Properties
- Non-hermetic packaging. Precise capacitance/impedance/prop

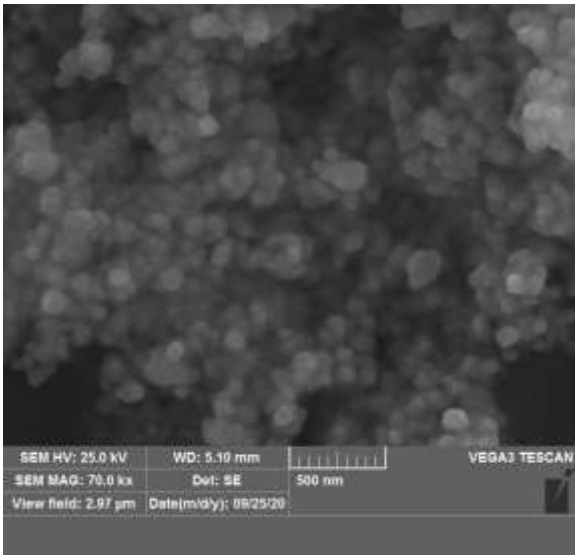
Applications

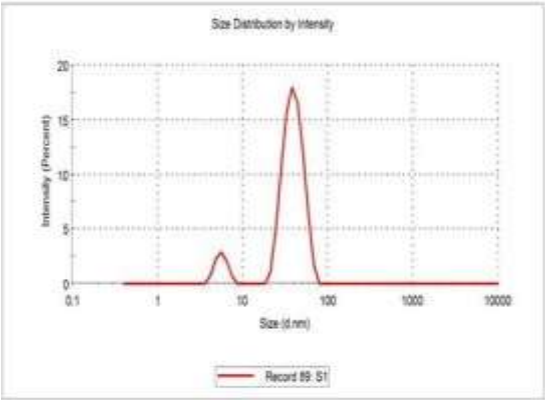
Cement Industries	Ceramic Coating	In The Adhesive And Sealant Field
Paints and Coatings	Used in Material Packaging	Biomedical Applications
Used in Paints and Inks	Used in Agriculture Industry	Used in Textile Industry

Specifications

SiO_2	Description
Purity	99.9 %
Average Particle Size	<100nm
Bulk Density	0.04G / Cm^3
Physical Form	Powder
Colour	White
Surface Area	~220m^2 / G
CAS No.	7631-86-9
Morphology Nano Particle	Near Spherical

SEM RESULTS





Titanium Dioxide Nanoparticles

We manufacture TiO2 nanoparticles by sol-gel method with multi levels of quality checks to obtain pure phase Titanium Dioxide nanoparticles.

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PRODUCT FEATURES

- Good Photocatalytic Ability
- Excellent Thermal stability
- Antimicrobial Properties
- Excellent UV blacking properties

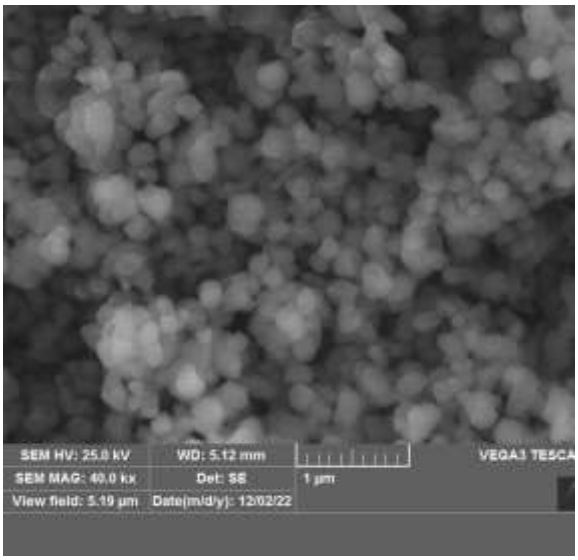
Applications

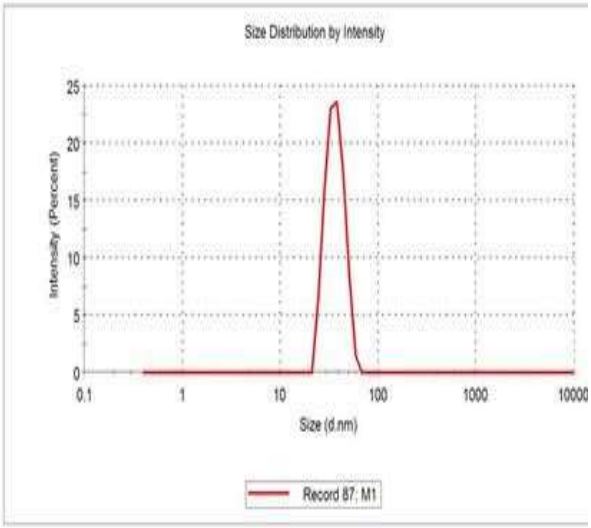
Polymer Nanocomposites	Ceramics	Photocatalyst
Paints and Coatings	Self-Compacting Concrete	Sensors

Specifications

TiO2	DESCRIPTION
Purity	99.9 %
Average Particle Size	<100nm
Thickness	1-5 Nm
Bulk Density	0.35 G / Cm3
Physical Form	Powder
Colour	White
Surface Area	~150m2 / G
CAS No .	13463 - 67-7
Morphology Nano Particle	Near Spherical

SEM RESULTS





Zinc Oxide Nanoparticles

We manufacture ZnO nanoparticles by sol-gel method in bulk quantity with multi levels of quality checks to obtain phase pure Zinc oxide Nanoparticles.

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APS: <100nm , Purity: > 99%

PRODUCT FEATURES

- Zinc oxide Nanoparticles are versatile
- Antibacterial, Antifungal and Antifouling agent
- Have good chemical stability, Thermal resistance
- Robustness and long shelf life
- Have selective Toxicity to Bacteria
- Have Excellent Biocompatibility and Low Toxicity
- Ability to Trigger excess reactive Oxygen species (ROS) production
- ZnO nanoparticles have UV-blocking properties

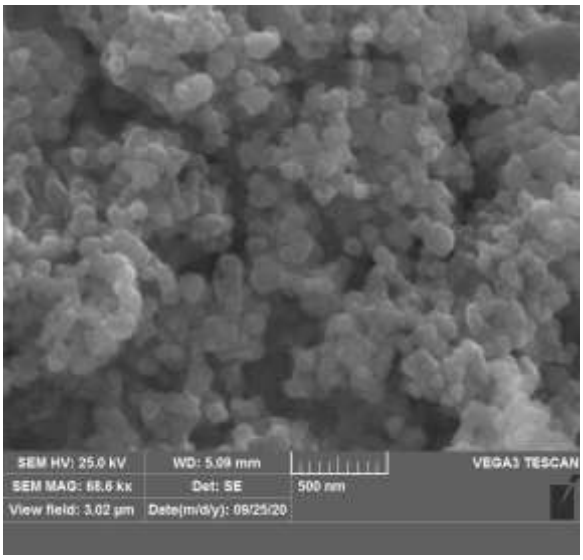
Applications

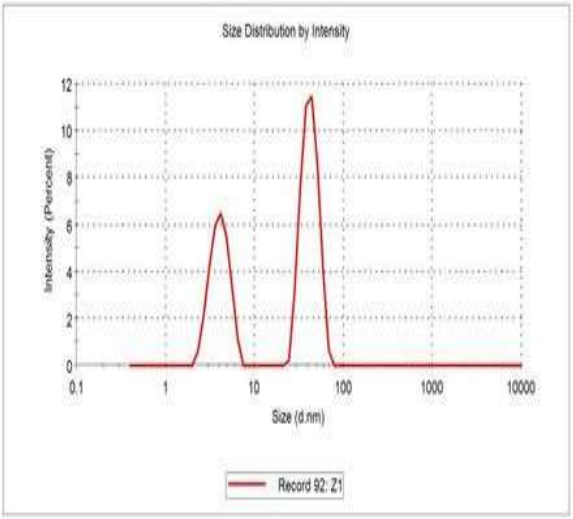
Used In Personal Care Products, Such As Cosmetics And Sunscreen	Excellent in Biomedical Applications, such As Anticancer, Drug Delivery	Antibacterial, and Diabetes Treatment, Anti-inflammation, Wound Healing And Bioimaging
Able To Improve the Flexural Strength Of self-compacting Concrete	Gas Sensor, Biosensor, Cosmetics, Storage, Optical Devices, Window Material	Paints, Coating, Fertilizers, Solar Cells

Specifications

Zinc Oxide Powder	Description
Purity	99.9 %
Particle Size	<100nm
Thickness	1-5 Nm
Bulk Density	0.58 G / Cm3
Physical Form	Powder
Colour	Black
Surface Area	~100-120m2 / G
CAS No. .	1314 - 13 -2

SEM RESULTS





CONDUCTIVE INK

Water Based Graphene Nano Platelets Ink (NADWGP-30)

NAD-WPG 30 is a water-based conductive ink that is easy to use on various substrates such as cotton, polyester, paper, non–woven fabric and electrodes. It can be used by multiple techniques including screen printing, roll to roll coatings, dip coating and brush coating.

Ad-Wpg30	Description
Sheet Resistance	**500 Ω/Sq Per 10 µm**
Surface Resistivity	**7.2 Milli Ω M**
Conductivity	**138.8 S/M**
Density	**1.58 G/Ml**
Stretchability	**2%**
Percentage Of Solid Loading	**30%**
Recommended Mesh	**100 Mesh**
Color	**Grayish Black**
CAS No..	**7782-42-5**
Shelf Life	**1 Months In Unopened Container**
Curing/ Sintering Temperature	**80 ° For 1 Minute** **60° For 5 Minutes**

Solvent Based Graphene Nano Platelets Conductive Ink (NADSNG-30)

AD-SGN 20 is a solvent-based conductive ink that is easy to use on various substrates such as polyamide, glass, paper, epoxy, electrodes and PET. It can be used by multiple techniques including screen printing, roll to roll coatings, dip coating and brush coating.

Ad-Sgn20	Description
Sheet Resistance	**500 Ω/Sq Per 10 µm**
Surface Resistivity	**4.6 Milli Ω M**
Conductivity	**217.19 S/M**
Density	**1.37 G/Ml**
Percentage Of Solid Loading	**20%**
Recommended Mesh	**100 Mesh**
Color	**Grayish Black**
CAS No..	**7782-42-5**
Shelf Life	**1 Month In Unopened Container**
Curing/ Sintering Temperature	**180° For 1 Minute** **100° For 5 Minutes**

Solvent Based Graphene Flakes Conductive Ink/Paste (NAD-SGR-20)

AD-SGR 20 is a solvent-based conductive ink that is easy to use on various substrates such as polyamide, glass, paper, epoxy, electrodes and PET. It can be used by techniques including screen printing, roll to roll coatings, dip coating and brush coating.

Ad-Sgn20	Description
Sheet Resistance	**350 Ω/Sq Per 10 µm**
Surface Resistivity	**3.82 Milli Ω M**
Conductivity	**261.8 S/M**
Density	**1.28 G/Ml**
Percentage Of Solid Loading	**2%**
Recommended Mesh	**100 Mesh**
Color	**Black**
CAS No..	**7782-42-5**
Shelf Life	**1 Month In Unopened Container**
Curing/ Sintering Temperature	**200° For 1 Minute** **180° For 3 Minutes**

DISCLAIMER: *The values are typical and are for general guidance and must not be used as a basis for specifications as concrete. Information contained in this publication, and otherwise supplied to users, is based on our general experience and is given in good faith, but we are unable to accept responsibility in respect of factors which are outside our knowledge or control. No warranty, either expressed or implied, is hereby made. The recommended industrial hygiene and safe handling procedures are believed to be generally applicable. Please refer to MSDS of respective product for safe use.*

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